

Reborn of the White Bone Demon: Role-Playing Game Design Using Generative AI in XR

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Figure 1: The gaming experience of users playing the role of White Bone Demon.

Abstract

This paper explores the application of Generative Artificial Intelligence (GenAI) techniques to the design of Role Playing Games (RPGs) in Extended Reality (XR) environments. We developed the game *Reborn of the White Bone Demon*, which utilizes AI speech emotion recognition technology to generate story lines and game assets in real-time based on the player's conversations with NPCs, enhancing the player's immersion and personalized experience, demonstrating the potential of GenAI in enhancing the XR gaming experience.

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1 INTRODUCTION

Role-playing games (RPGs) emphasize players creating their own identities and experiencing the game story from the perspective of such identities [Niarchos et al. 2023], and more and more RPGs are adopting immersive XR environments to increase the sense of spatial embodiment, however, the almost fixed character dialogues and asset contents diminish the player's sense of immersion and personalization of their experience in the game. Currently, the development of GenAI brings richer experiential possibilities to XR games [Sun et al. 2023], which can inspire the potential of RPGs in terms of real-time generation of game text, character dialogues, image mapping, and 3D props. Based on this, we designed and developed a game, *Reborn of the White Bone Demon* (figure 1), to explore the possibilities of GenAI-enabled RPG experiences in immersive XR environments. The game is deployed based on Meta Quest3, which allows one player to participate for 15 minutes at a time with unlimited activity space.

Our contributions:

- In terms of character dialog input, we use AI speech emotion recognition to detect the emotional input of the player's

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dialog and generate real-time dialog feedback from the non-player characters (NPCs) to enhance the immersive feeling of the RPG;

- In terms of game asset output, we use GenAI to generate real-time game storylines, environment mapping, and interactive props, enhancing the immersion of the RPG game.

2 DESIGN AND IMPLEMENTATION

We based the design of the game on the classic story *Three Strikes Against the White Bone Demon* from the ancient Chinese novel *Journey to the West*. In this game, the user is required to role-play a demon and drive the narrative development of the game by conversing with the three characters: Tang Monk, the Sha Monk, and Zhu Bajie as AI agents. Throughout the game, the GenAI will also advance the story based on the content and emotions of the user's conversations with the NPCs, and generate 3D model props and scenes in real time to enhance the immersion of the game.

2.1 Interaction Design

Customized Language Model. We trained a dialog model based on the original text of the classic story, character traits, etc., and asked the character NPC of the AI agent to gradually guide the player to utter preset keywords. The answer contains the NPC's look and hints. Based on this, the system will generate corresponding plot settings (e.g. The young girl claims that her village has been attacked by a demon and pleads for help.) and unknown target prop keywords (e.g. Brother's jade pendant) based on the user's selected identity and strategy cards. The keyword of the target prop (e.g. the brother's jade pendant) is unknown. The user has to play the role of this character throughout the game and utilize the corresponding strategies to gradually guess and trigger the keyword based on the conversation with the NPC character to win the game.

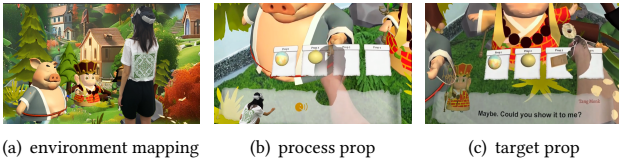


Figure 2: Real-time generation according to the description.

Interaction Process Design. The game is divided into two phases: scene generation and prop generation. After entering, the user will first be guided to describe a scene (e.g. the village of the girl) (figure 2-a), and this description will generate real-time scene to enhance the user's sense of immersion. In the prop generation phase, the user can continuously talk with NPCs, and the system will generate corresponding 3D model props (e.g. a bun for Zhu Bajie) (figure 2-b) in real time according to the props words in the user's speech, based on which the user can interact with NPCs. In this process, the user needs to continuously have a sincere and emotional voice dialog with the NPCs to enhance the NPCs' trust in the character played by the user to get positive feedback about the target prop keywords (e.g. brother's jade pendant) (figure 2-c).

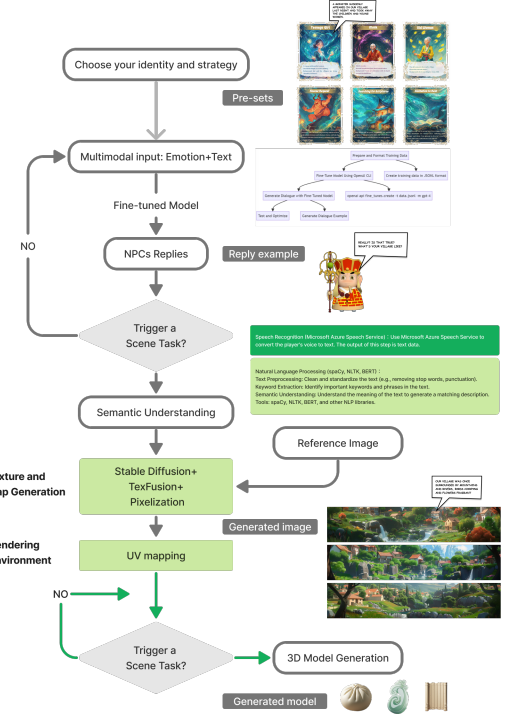


Figure 3: Technical workflow

2.2 AI Generation

GenAI is used throughout the interaction process, and the speech text of each conversation will be used to generate content and adjust narrative development by recognizing emotions and keywords of objects that can be generated in the text (figure 3).

AI recognizes text and emotions to advance story development. The system will call different NPCs to give feedback based on the user's voice text, which in turn will advance the story based on the plot setting. In this process, the user needs to engage in voice conversations with NPCs sincerely and with emotions, so as to obtain feedback that is more closer to the keyword prompts.

AI recognizes text to generate props and scenes. Based on the user's description, the system will generate real-time scene mapping and 3D model props based on spatial positioning.

3 CONCLUSION

In this paper, we explore the application model and potential of GenAI for role-playing games in XR environments, and enhance the playability of traditional game stories and established scripts in terms of both character dialog inputs and game asset outputs, and improve the player's sense of immersion and participation in role-playing, and we hope that this work can provide a way of thinking for the future combination of GenAI and XR games.

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